SCHEMATIC — errors exaggerated; dashed = design. Not measured data.

(1) gap error: one gap narrower than designed

→ that coupling κ grows (log-normal coupling disorder; keeps chiral symmetry)

(2) width error: centre guide wider (centre position fixed by mask)

→ its β grows AND both neighbouring gaps shrink (correlated width model; breaks chiral symmetry)

(3) wafer-scale thickness error: every guide thicker

→ every β shifts together (global shift; cancels in a relative-phase measurement)